

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

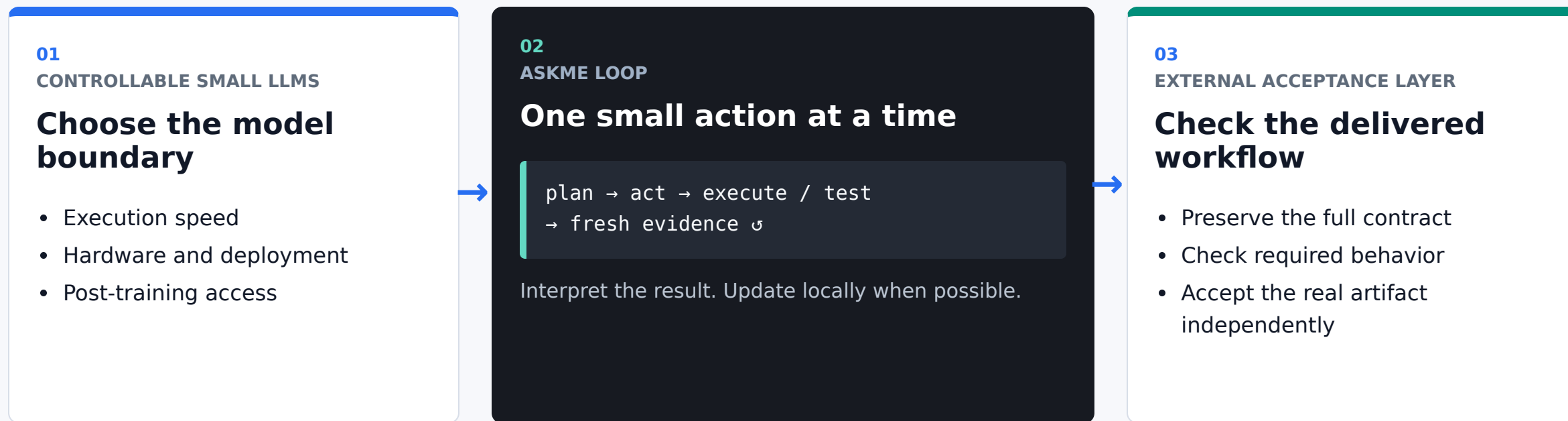
Denis Akhyyarov

Sr Staff Research Scientist at ServiceNow

[@den-run-ai](#) · github.com/den-run-ai/askme

AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one native tool call → execution evidence ↻



The bridge: eight native tools + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c  
build ./main  
run ./main  
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

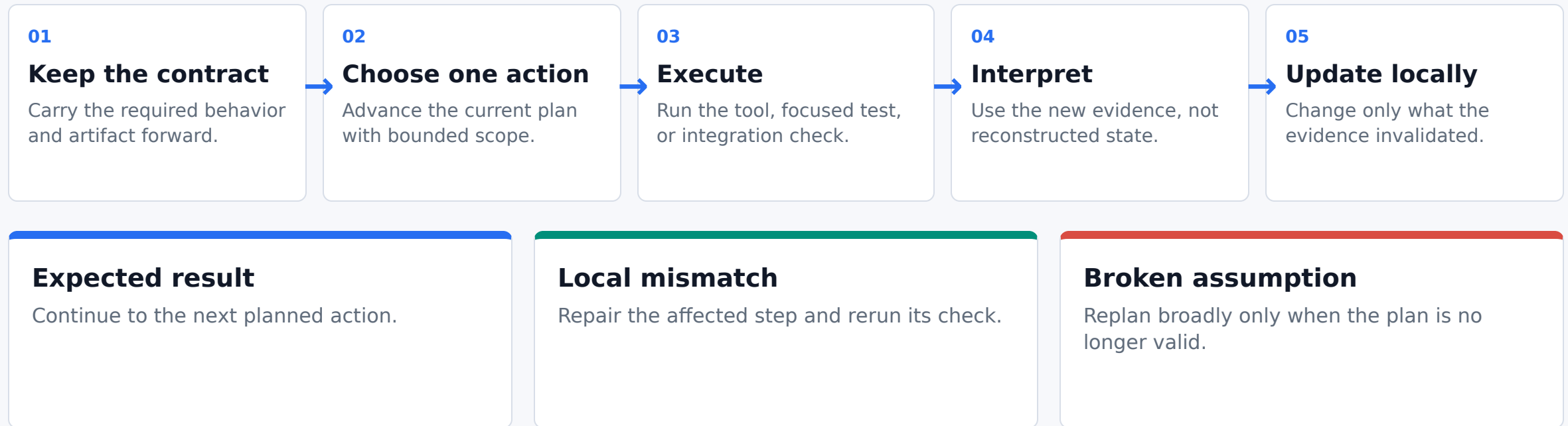
From the agent's side, the task looked done.

Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion $\neq$ accepted artifact.

Hypothesis: update only what fresh evidence invalidates



Design goal: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

Acceptance caught the one bad deliverable

Four hosted models · two simple tasks · one run each

8 / 8

reported complete

7 / 8

artifacts accepted

The fastest build was the rejected one.

Completion and speed were not enough.

HOSTED VARIANT	ARTIFACT BUILD	SANITY REPAIR
Gemma 4 26B A4B	Accepted	Accepted
Gemma 4 31B	Accepted	Accepted
Qwen3.6-27B	Accepted	Accepted
Qwen3.6-35B-A3B	Wrong path	Accepted

Compatibility smoke, not a ranking: single runs on simple tasks — no model comparisons.

Applied patches, unresolved feature task

After bundled harness changes and a changed serving stack.

BEFORE · JULY

0 writes

Empty patch

Revision 2 left both cells empty. Their failure modes differed.

AFTER · AUG 1

Patches applied

→ **11 / 13** **7 / 13**

Gemma target tests Qwen target tests
Held-out target tests passed in part;
both patches remained unresolved.

GAPS IN BOTH ATTEMPTS

No target-test run

→ Gemma rewrote one file 18 times.
Qwen returned to reading. Both agents
exhausted their planning attempts.

Observed: partial target-test acceptance, without agent completion.

One task, one attempt per model · historical results · no causal attribution or benchmark score.

Local repairs are possible. Reliable autonomy is unproven.

Observed locally

Gemma 4 E4B (dense PLE), Aug 4: four repairs of one seeded health-check bug passed independent acceptance. All four agents exhausted.

Supported

Narrow, independently accepted small repairs exist. A correct artifact is not a clean agent finish.

Still open

Dependable autonomous coding, net time savings, and causal harness benefit. The historical pilots do not isolate a harness effect.

Evaluate the model, harness, and task as one system.

One seeded task—not a general readiness verdict or time-savings claim.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

AskMe and pi

How the harness divides work between the model and controller.

	AskMe	pi
ACTION SURFACE	8 native tools: 6 executable actions + <code>done</code> / <code>fail</code> . One call per turn.	4 default tools: read, write, edit, bash. Extensions can change the tool set.
STATE + CONTROL	Explicit plan, curated slim state, bounded replanning.	Model-led session tree with branching and compaction.
COMPLETION BOUNDARY	<code>done</code> + conditional validation. Unavailable check: <code>complete_unverified</code> .	Loop ends when tool calls stop. Acceptance comes from an external check.

Historical feature probe · August 1, 2026 · target tests passed

Dense model	AskMe revision 3	pi archival attempt
Gemma 4 31B	11/13 · unresolved	11/13 · unresolved
Qwen3.6-27B	7/13 · unresolved	10/13 · unresolved

One task, one attempt per cell. Unmatched providers and budgets; unqualified exploratory evidence, not a current performance ranking.